An AI-assisted investigation of a Spin(10) × global $U(1)_{PQ}$ gauge/scalar candidate

Construction, bounded results, and failed tests

Solo research discussion note, prepared with AI assistance. Version 2.

Abstract

This note defines a nonsupersymmetric, renormalizable gauge/scalar construction and summarizes selected calculations already recorded for it. A complete scalar-invariant basis and a locally positive high-scale scalar point have been obtained. An initial one-light-Higgs tuning fails its heavy-relaxed tree-level quartic test; a later distinct point has a positive relaxed quartic. The recorded gauge-matching equations also retain a nonzero mismatch; model-specific finite two-loop heavy thresholds remain incomplete. Selected classical symmetry identities and one-loop scalar operator closure have been checked within stated scopes. These results neither establish a physically viable grand unified theory nor exclude the entire construction. This is a technical discussion draft, not a new scientific acceptance decision.

1. Assumptions and fields

Work in four spacetime dimensions with gauge group Spin(10) and global PQ selection rules. The simply connected group permits the fermionic spinor 16; local tensor calculations use the $\mathfrak{so}(10)$ action. A possible quotient by common gauge/PQ centers has not been fixed. No supersymmetry, explicit soft PQ violation, or additional scalar representation is assumed.

Field	Representation/type	q_{PQ}	Real scalar coordinates
Φ_{ij}	real symmetric traceless 54	0	54
Σ_{ijklm}	complex self-dual 126	+2	252
ϕ_i	complex 10	−2	20
S	complex singlet	−4	2
ψ_p, $p = 1, 2, 3$	left-handed 16_F	+1	—

Indices run from 0 to 9, with Euclidean vector metric δ_{ij} and $\epsilon_{0\ldots9} = +1$. Fix $*\Sigma = +i\Sigma$; its conjugate is the $\overline{126}$. The scalar kinetic convention is

$$\mathcal{L}_{\mathrm{kin}} = \frac{1}{2}(D_\mu\Phi_{ij})^*(D^\mu\Phi_{ij}) + \frac{1}{2\,5!}(D_\mu\Sigma_{ijklm})^*(D^\mu\Sigma_{ijklm}) + (D_\mu\phi_i)^*(D^\mu\phi_i) + (\partial_\mu S)^*(\partial^\mu S).$$

The self-dual factor avoids double counting complementary five-index sectors. The physical construction uses the usual gauge-curvature kinetic term and $\mathcal{L} = \mathcal{L}_{\mathrm{gauge}} + \mathcal{L}_{\mathrm{kin}} + \mathcal{L}_Y - V$. The allowed symbolic Yukawa structures are $\psi\psi\phi$ and $\psi\psi\Sigma^*$, with complex symmetric family matrices and isometrically normalized intertwiners. Their numerical matrices and right-handed-neutrino spectrum have not been selected.

The field-content motivation follows Babu and Khan [1]; PQ/axion context is discussed in [2,3]. The potential below is a separately specified source-derived variant. Published scalar benchmarks, fermion fits, and unification conclusions are not imported as results for it. No novelty claim is made.

2. Complete frozen scalar definition

Write $V = V_2 + V_3 + V_4$. Repeated tensor indices are summed over all ten values. Define

$$N = \frac{1}{5!}\Sigma_{abcde}\Sigma^*_{abcde}, \quad u = \phi_i^*\phi_i, \quad t = \phi_i\phi_i, \quad p_n = \mathrm{Tr}(\Phi^n),\ n = 2,3,4,$$

$$K_{ij} = \frac{1}{4!}\Sigma_{iabcd}\Sigma^*_{jabcd}, \qquad T_{ij} = \frac{1}{4!}\Sigma_{iabcd}\Sigma_{jabcd},$$

$$L_{\Phi\Sigma} = \frac{1}{2\,3!}\Phi_{ij}\Phi_{kl}\Sigma_{ikabc}\Sigma^*_{jlabc}, \qquad E = \frac{1}{2!\,2!\,3!}\Sigma_{ijlmn}\Sigma_{ijpqr}\Sigma^*_{lmpqr}\phi_n.$$

T is symmetric traceless. For increasing tuples I, J of length $5-k$, let $P_k[I,J] = \sum_{K\in\binom{\{0,\ldots,9\}}{k}} \Sigma_{KI}\Sigma_{KJ}$ and $Q_k = \sum_{I,J}|P_k[I,J]|^2$, for $k = 0,1,2$. Antisymmetric components with repeated indices vanish. If A, B are disjoint, $\varepsilon(A,B)$ is the parity of their concatenation relative to increasing order; otherwise it is zero. The fourth real five-form quartic is

$$X_{131} = \sum_{I,J\in\binom{\{0,\ldots,9\}}{4}} \sum_{\substack{A\subset I,\ |A|=3\\ B\subset J,\ |B|=1}} \varepsilon(A, I\setminus A)\varepsilon(B, J\setminus B)\varepsilon(A,B)\varepsilon(I\setminus A, J\setminus B)$$
$$\times P_1[I,J]\,\overline{P_1[\mathrm{sort}(A\cup B), \mathrm{sort}((I\setminus A)\cup(J\setminus B))]}.$$

The potential in these normalizations is

$$\begin{aligned}
V_2 &= m_\Phi^2 p_2 + m_\Sigma^2 N + m_\phi^2 u + m_S^2|S|^2,\\
V_3 &= \mu_\Phi p_3 + \mu_{\Phi\phi}\Phi_{ij}\phi_i^*\phi_j + [z_6 t S^* + \text{h.c.}],\\
V_4 &= \lambda_{\Phi1}p_2^2 + \lambda_{\Phi2}p_4 + \lambda_{\Phi\Sigma1}p_2 N + \lambda_{\Phi\Sigma2}L_{\Phi\Sigma}\\
&\quad + \lambda_{\Phi\phi1}p_2 u + \lambda_{\Phi\phi2}\phi_i^*\Phi_{ij}\Phi_{jk}\phi_k + \lambda_{\Phi S}p_2|S|^2\\
&\quad + \lambda_{\Sigma1}Q_0 + \lambda_{\Sigma2}Q_1 + \lambda_{\Sigma3}Q_2 + \lambda_{\Sigma4}X_{131}\\
&\quad + \lambda_{\Sigma\phi1}N u + \lambda_{\Sigma\phi2}\phi_i^*K_{ij}\phi_j + \lambda_{\phi1}u^2 + \lambda_{\phi2}|t|^2\\
&\quad + \lambda_{\Sigma S}N|S|^2 + \lambda_{\phi S}u|S|^2 + \lambda_S|S|^4\\
&\quad + [z_4\Phi_{ij}T_{ij}S + z_K\Phi_{ij}\phi_i\phi_j S^* + z_\eta E + z_D T_{ij}\phi_i\phi_j + \text{h.c.}].
\end{aligned}$$

All m^2, μ, λ coefficients are real; $z_6, z_4, z_K, z_\eta, z_D$ are complex. The masses have dimension two, the cubics dimension one, and the quartics dimension zero. Hermitian conjugation includes the coefficient. The shorthand $\mu_{\Phi\phi}, \lambda_{\phi1,2}, \lambda_{\phi S}, z_\eta$ corresponds to repository names `muPhiPhi`, `lambdaPhiVector1,2`, `lambdaVectorS`, and `zEta`.

There are 29 Hermitian coefficient groups: four quadratic, three cubic, and 22 quartic. Their 24 real and five complex coefficients give 34 real directions before rephasings. Independent D_5 character multiplicities give 34 complex singlet slots before Hermitian pairing; explicit contractions attain that upper bound. The generic scalar phase quotient leaves three relative coupling phases. This basis result fixes the action, not a viable vacuum.

3. Scalar results: a failed tuning and a later positive point

The candidate high-scale background is

$$\langle\Phi\rangle = \frac{\omega}{\sqrt{60}}\,\mathrm{diag}(-2\mathbf{1}_6, 3\mathbf{1}_4), \quad \langle\Sigma\rangle = \frac{\sigma}{4\sqrt{2}}\bigwedge_{k=0}^{4}(e_{2k+1}+ie_{2k}), \quad \langle S\rangle = \frac{v}{\sqrt{2}}, \quad \langle\phi\rangle = 0.$$

The first direction has the intended Pati–Salam/discrete-parity stabilizer; the combined gauge orbit has 33 broken and 12 unbroken directions, consistent with the SM Lie algebra. Global topology and the PQ quotient are separate questions. Tadpoles and Hessians are derived from the same frozen potential.

At the recorded Stage 1 point, $\omega = \sqrt{60}$ in calculation units and $\sigma/\omega = v/\omega = 0.1$. The exact-decimal free coefficients and tadpole-eliminated masses are specified in `STAGE1_POINT.json` and its replay program. All 35 SM-irrep multiplicity blocks were evaluated. The real Hessian has rank 294 and nullity 34: 33 gauge directions plus one PQ direction. The remaining eigenvalues are positive in the canonical kinetic metric; the smallest recorded physical mass squared is approximately 0.08656. This is local positivity modulo symmetry, not global boundedness or a phenomenologically fitted scale. This point has no light Higgs doublet.

The distinct Stage 2 point changes $\mathrm{Re}\,z_6, \mathrm{Re}\,z_K, \mathrm{Re}\,z_\eta$ to 0.3 in the stored dimensionless convention and tunes $m_\phi^2/\omega^2 \simeq -0.1337864654$ to the exact algebraic root recorded in `STAGE2_POINT.json`. It has rank 290 and nullity 38, including four real coordinates of one massless complex SM Higgs doublet. All physical heavy quadratic modes remain positive. This statement alone is insufficient for local stability along the tuned flat direction.

Decisive additional test

For a canonically normalized neutral field h, $H^\dagger H = h^2/2$, expand around the tuned point with physical heavy coordinates X_A:

$$V = V_0 + a_{\mathrm{direct}}h^4 + h^2 j_A X_A + \tfrac{1}{2}X_A\mathcal{H}_{AB}X_B + \cdots .$$

Minimization on the positive heavy subspace gives $X = -\mathcal{H}^+ j h^2 + \cdots$, and hence

$$\lambda_{\mathrm{eff}} = 4\left(a_{\mathrm{direct}} - \tfrac{1}{2}j^T\mathcal{H}^+ j\right), \qquad V_{\mathrm{eff}} = \lambda_{\mathrm{eff}}(H^\dagger H)^2 + \cdots .$$

The recorded parent-action evaluation gives

Quantity in $(H^\dagger H)^2$ normalization	Value
Fixed-heavy direct quartic	+0.19910
Heavy relaxation subtraction	56.68893
Relaxed effective quartic	**−56.48984**

The negative sign makes the Stage 2 point a failed local scalar minimum at tree level. It does not invalidate the separate Stage 1 result or prove that every parameter point fails. A direct parent-potential expansion along the heavy-field valley reproduces the quartic; heavy-coordinate shear and complex Higgs-phase controls preserve it. These extraction routes share the frozen tensor implementation. No global-minimum or loop-corrected stabilization claim is made.

A distinct positive-quartic point

The later recorded point in `canonical_so10_positive_higgs/POINT.json` retains the Stage 1 VEVs but uses $z_6/\omega = z_K = 0.03$, $z_\eta = 0.3$, $\lambda_{\Phi\phi 1} = -0.002$ and an exact algebraic tuning $m_\phi^2/\omega^2 \simeq 0.007302047861$. Its real Hessian has rank 290 and nullity 38, with positive heavy modes. The direct quartic is 0.19845235; heavy relaxation subtracts 0.09359411, leaving

$$\boxed{\lambda_{\mathrm{eff}} = +0.10485824.}$$

The point's SHA-256 is `476645438ea8a7ada42fea90e80174b772d45ce3e729d013d134e9848950c816`. This is a local tree-level scalar benchmark, not global boundedness or a viable GUT. The documented amendment changes a mixed coupling and compensates its quadratic mass contribution, preserving the Higgs tuning while reducing its heavy-singlet source. Direct-parent replay and a valley-polynomial extraction check the positive quartic with shared tensor inputs. The failed earlier point remains failed; this new witness excludes a universal claim that all such Higgs tunings must have nonpositive quartics.

4. Gauge matching: a measured mismatch, an incomplete verdict

The recorded direct one-loop calculation matches the full parent to the SM in a common $\overline{\mathrm{MS}}$ background-field Feynman-gauge convention, with GUT-normalized g_1. It includes 290 heavy real scalar directions and 33 massive vector/Goldstone/ghost sets from the later positive-quartic point, not the earlier failed tuning. The threshold ledger and matching program bind that point's hash explicitly. Global boundedness remains unresolved.

Define $\alpha_i = g_i^2/(4\pi)$ and one-loop threshold functions λ_i by

$$\alpha_i^{-1}(\mu) = \alpha_{10}^{-1}(\mu) - \frac{\lambda_i(\mu)}{12\pi}.$$

The complete state accounting gives

$$b_{\mathrm{SM}} = \left(\frac{41}{10}, -\frac{19}{6}, -7\right), \quad b_S = \left(\frac{377}{30}, \frac{77}{6}, \frac{79}{6}\right), \quad T_V = (8, 6, 5),$$
$$b_{\mathrm{SM},i} + b_{S,i} - \frac{7}{2}T_{V,i} = -\frac{34}{3} = b_{10}, \qquad \frac{d\lambda_i}{d\ln\mu} = 6(b_{\mathrm{SM},i} - b_{10}).$$

These are exact beta-jump and one-loop matching-scale checks. At $\mu = \omega$, the frozen inverse-coupling inputs are $(59.0100, 29.5800, 8.47458)$ at M_Z. The one-loop equations are

$$\alpha_i^{-1}(M_Z) = \frac{4\pi}{g_{10}^2} - \frac{\lambda_i^S + \lambda_i^V(g_{10})}{12\pi} + \frac{b_i^{\mathrm{SM}}}{2\pi}\ln\frac{\omega}{M_Z}.$$

They have no common perturbative solution for the frozen inputs. The recorded least-squares residuals, in inverse-coupling units and channel order $(1, 2, 3)$, are:

Completed approximation	r_1	r_2	r_3	$\max\lvert r_i\rvert$
One-loop running and thresholds	0.36615	−1.05980	0.69389	1.05980
Two-loop SM running, one-loop thresholds	0.26091	−0.74548	0.48479	0.74548

The latter uses the SM running framework [5] and frozen reference boundaries from [6], including top, bottom, and tau insertions with their one-loop evolution. A separate inverse-coupling evolution with a different adaptive integrator reproduces the residual to better than 2×10^{-8} under the frozen equations. Matching-scale variation $\mu/\omega \in [0.5, 2]$ leaves maximum residuals from 0.74172 to 0.74931.

What the missing calculation must supply

The exact two-loop RG logarithmic template has been derived, but its finite model-dependent heavy constants have not. The published general matching framework [4] provides methodology and a restricted comparator, not the finite answer for these mixed, nondegenerate fields and multiple VEVs. The complete Yukawa-dependent threshold also requires numerical family matrices and right-handed-neutrino masses.

Small scale variation is not a rigorous bound on unknown finite constants. Setting them to zero or freely tuning them would not compute a prediction. Consequently the formal gauge verdict remains unresolved: neither unification success nor exclusion has been established. The diagnostic fitted $g_{10} \simeq 0.5231$ and $\omega \simeq 1.64 \times 10^{15}\,\mathrm{GeV}$ are not physical benchmark parameters. No replacement spectrum or flavor fit is introduced in this note.

5. Symmetry and quantum evidence: scope matters

A separately named covariant physical gauge/scalar candidate satisfies exact classical local gauge identities through factorized representation checks. The recorded primary and tensor-functor replay compare the 328-real scalar representation action and all 45 generator directions. The usual $\delta(Dx) = -g_{10}\rho(\omega)Dx$ identity, invariant kinetic metric, curvature identity, and invariant potential establish the classical statement. This qualification is inactive; it does not replace a promoted component AST or complete a quantum calculation.

Earlier literal component-action failures are retained. One preserved two-field witness has nonzero gauge variation

$$\delta\mathcal{L}_{\text{old}} = \frac{ig_{10}}{\sqrt{2}}(\partial_\mu C^4)(Y\partial^\mu X - X\partial^\mu Y),$$

with nonzero ghost Euler derivative. The later covariant candidate fixes the kinetic structure by derivation. A failed encoded candidate is evidence about that representation of the action, not a theorem against the gauge group.

Separately, the recorded one-loop parent scalar-pole projection has exact rank $4 + 4 + 26 = 34$ and zero out-of-basis residual in its tested operator space. Hermiticity and PQ selection-rule audits also pass within that scope. This bounded radiative-closure result is not a complete anomaly or quantum consistency theorem; it does not evaluate finite two-loop thresholds.

In the gauge-fixing/source investigation, a preserved normalization candidate misses a selected auxiliary–antighost–ghost cubic pole. At one recorded real-component witness, in common $\hbar g_{10}\xi^2/(32\pi^2\epsilon)$ units, the coefficients are

$$G_{\text{actual}} = \frac{48\sqrt{10}}{25}, \quad G_{\text{candidate}} = \frac{8\sqrt{10}}{5}, \quad G_{\text{actual}} - G_{\text{candidate}} = \frac{8\sqrt{10}}{25} \neq 0.$$

Later bounded restoration investigations do not automatically erase this failed candidate or establish full quantum consistency. The ledger remains at 34 of 39 executed counterterm items, with the relevant quantum gate blocked. New uncommitted vector/ghost work was undergoing qualification at recovery and is excluded from this note's earned claims. No completion is inferred from a handoff or a live worker.

6. Remaining obligations and questions

The later positive-quartic point establishes a bounded local tree-level scalar witness. Physical viability still requires controlled interactions and loop corrections; global boundedness and vacuum competition; a complete consistent quantum gauge/source normalization, including measure, anomaly, and infrared obligations; finite heavy-threshold matching; a fitted matter/neutrino sector; and derived phenomenology. PQ anomaly coefficients, the physical axion, global quotient, and topology also require their own analysis. The presence of PQ selection rules alone does not establish an axion prediction or a strong-CP solution for this variant. No experimental agreement is claimed.

Two useful discussion questions are: which analytic constraints on $j^T\mathcal{H}^+j$ best screen other scalar points after the failed Higgs tuning; and which controlled finite-threshold calculation or bound is needed before assessing the frozen gauge mismatch? These questions do not presume that missing corrections will rescue the model.

AI assistance and independence

LLMs assisted calculation development, code, verification work, and preparation of this note. Scientific direction and acceptance belong to the human author. Distinct derivations and negative controls are useful, but shared action inputs, some tensor/arithmetic code, Python/SymPy, and analyst limit independence. Selected numerical work also uses NumPy, SciPy, mpmath, CVXPY/Clarabel; selected tensor checks have a Julia replay. No external scientist's verification, proof-assistant certification of the whole model, or endorsement is claimed.

7. Evidence and reproduction map

This note synthesizes recorded calculations. The reproduction guide separately records package-byte checks and short read-only replays performed for publication. Its input snapshot is local Git commit

`8a07e12f61350e1c6f488d0c52d7d23c676e156c`.

The source scientific authority is `project_state.json`, version 113. Public manifests authenticate exported bytes without promoting that ledger. Historical custody remains unresolved, not a global pass.

Evidence paths below are relative to the source snapshot. The original Git history is not included. The reproduction guide provides the packaged inputs, scripts, dependencies, and evidence records for the note's claims.

Claim	Evidence locator
Scalar definition	`calculations/canonical_so10_scalar_reconstruction/` `PARENT_ACTION_V1.md`, `RESULT_V2.md`, `RECORD.md`
Hessian and points	`calculations/canonical_so10_full_hessian/RESULT.md` `calculations/canonical_so10_scalar_benchmark/RESULT.md`, `STAGE1_POINT.json`, `STAGE2_POINT.json`
Failed / positive quartics	`calculations/canonical_so10_light_higgs_quartic/RESULT.md` `calculations/canonical_so10_positive_higgs/RESULT.md`, `POINT.json`
Gauge residuals	`calculations/canonical_so10_direct_gauge_matching/RESULT.md` `calculations/canonical_so10_direct_gauge_next_order/RESULT.md`, `next_order.json`
Missing finite boundary	`calculations/canonical_so10_direct_two_loop_heavy_threshold/` `RESULT.md`, `YUKAWA_CONVENTION_V1.md`
Scalar pole closure	`tools/two_loop_qft_compiler_v1/LAYER5B_M05_M08_RESULT.md`
Classical and normalization	Under `tools/two_loop_qft_compiler_v1/` `m08_independent_replay_package_v1/`: `A10_PHYSICAL_ACTION_RECONCILIATION_RESULT.md`, `A10_E5_AUXILIARY_VERTEX_RESULT.md`

For targeted reproduction, the result files specify full prerequisites and controls. From an isolated snapshot, representative entry points include

```
python -B calculations/canonical_so10_scalar_reconstruction/verify_action_inventory.py
python -B calculations/canonical_so10_scalar_benchmark/replay_candidate_exact.py
python -B calculations/canonical_so10_scalar_benchmark/replay_stage2_exact.py
python -B calculations/canonical_so10_light_higgs_quartic/compute.py
python -B calculations/canonical_so10_positive_higgs/verify_exact_point.py
python -B calculations/canonical_so10_direct_gauge_next_order/independent_replay.py
```

Some calculation scripts write outputs. They should not run in a checkout with an active immutable-input validation. The full Hessian, quartic, and quantum-pole calculations were not rerun for publication, and no new parameter search was performed.

References

[1] K. S. Babu and S. Khan, *A Minimal Non-Supersymmetric SO(10) Model: Gauge Coupling Unification, Proton Decay and Fermion masses*, Phys. Rev. D **92**, 075018 (2015). arXiv:1507.06712.

[2] R. D. Peccei and H. R. Quinn, *CP Conservation in the Presence of Pseudoparticles*, Phys. Rev. Lett. **38**, 1440 (1977). doi:10.1103/PhysRevLett.38.1440.

[3] A. Ernst, A. Ringwald and C. Tamarit, *Axion Predictions in SO(10) × U(1)$_{\rm PQ}$ Models*, JHEP **02** (2018) 103. arXiv:1801.04906.

[4] W. Martens, *Towards a Two-Loop Matching of Gauge Couplings in Grand Unified Theories*, JHEP **01** (2011) 104. arXiv:1011.2927.

[5] M. Luo and Y. Xiao, *Two-loop Renormalization Group Equations in the Standard Model*, Phys. Rev. Lett. **90**, 011601 (2003). arXiv:hep-ph/0207271.

[6] D. Buttazzo et al., *Investigating the near-criticality of the Higgs boson*, JHEP **12** (2013) 089. arXiv:1307.3536.